

Figures 1 and 2 illustrate the following general fact about linear systems, to be verified in Section 1.2.

A system of linear equations has 1. No solution, or 2. exactly one solution, or 3. infinitely many solutions

A system of linear equations is said to be consistent if it has either one solution or infinitely many solutions; a system is inconsistent if it has no solution.

Matrix Notation

Given the system

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ 5x_1 - 5x_3 = 10 \end{cases} \quad (3)$$

with the coefficients of each variable aligned in columns, the matrix

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$$

is called the coefficient matrix (or matrix of coefficients) of the system (3), and

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix} \quad (4)$$

is called the augmented matrix of the system.

An augmented matrix of a system consists of the coefficient matrix with an added column containing the constants from the right sides of the equation. If m and n are positive integers, an $m \times n$ matrix is a rectangular array of numbers with m rows and n columns.

Solving a Linear System

This section and the next describe an algorithm, or a systematic procedure, for solving linear systems. The basic strategy is to replace one system with an equivalent system (i.e. one with the same solution set) that is easier to solve.